Riverline Learn Hub: Extended Curriculum Research & Architectural Blueprint

Integrating Epistemic Confidence Tiers, 30–60 BB Rake-Free Home Game Geometry, 9-Max 100 BB+ Online Cash Strategy, Personal Strategy Alignment, and a 58-Concept Dependency Map

Independent Research Deliverable · Riverline Workstation (M2 → M3)

This research specification synthesizes foundational No-Limit Hold'em literature, coaching curricula, solver licensing audits, and the architectural rules of `RIVERLINE_BRIEF.md` (2026-10-21) into a unified design and curriculum reference for Riverline's Learn Hub.

- **Complete 58-Concept Dependency Map (#1–#58):** Expands the initial 50-concept cash game path with 8 dedicated Riverline concepts covering **30–60 BB rake-free home games**, **9-max 100 BB+ online full-ring**, **Incomplete Hand Reconstruction (Unknown preservation)**, **Personal Strategy vs. Baseline Mix comparison**, and **Epistemic Confidence Literacy**.
- **Four-Tier Epistemic Confidence Mapping:** Replaces binary "Exact vs. Judgement" tags with both cognitive modes and Riverline's strict provenance badges: `Exact (scoped)`, `Verified (solver/combinatorial benchmark)`, `Estimated (heuristic baseline)`, and `Unavailable (unmodeled state)`.
- **Canonical Shared Domain Ownership (`shared/poker-domain/`):** Maps every quantitative formula (Pot Odds, SPR, Combinatorics, Hand-vs-Range Equity, MDF, Geometric Sizing, Variance) to `shared/poker-domain/` so the UI layer only formats outputs.
- **Re-Engineered First 8 Core Onboarding Lessons:** Dual-track interactive lessons contrasting 40 BB rake-free home games with 100 BB+ raked online games, obeying "calm surface, deep data" and one primary action per view.
- **Trilingual Localization (EN, RU, HE) & Solver Licensing Audit:** Standardized terminology and RTL/LTR token isolation (`<bdi dir="ltr">`) plus direct verification of b-inary/postflop-solver (AGPL-3.0) and the HoldemResources Calculator (HRC v4) EULA.

Document Metadata	Specification	Core Architectural Scope	Coverage
Project & Target	Riverline Learn Hub (M2 Rework → M3 Expansion)	**Curriculum Breadth**	58 Concepts across 6 Tiers + First 8 Core Lessons
Primary Game Profiles	6–8 Handed 30–60 BB No-Rake Home & 9-Max 100 BB+ Online	**Epistemic Tiers**	Exact (Scoped), Verified, Estimated, Unavailable
Supported Locales	English (EN), Russian (RU), Hebrew (HE — RTL + LTR tokens)	**Code Boundary**	`shared/poker-domain/` vs. UI Formatting Layer

Executive Synthesis: How the Riverline Project Brief Transforms the Research

Integrating `RIVERLINE_BRIEF.md` (dated 2026-10-21) directly upgrades the Learn Hub research from a standalone poker syllabus into a tightly coupled educational specification for a **No-Limit Texas Hold'em study workstation** ("the honest poker coach for the hands you actually play"). Rather than treating lessons as isolated textbook pages, the brief mandates that every concept connect hand reconstruction, replay and review, mathematical analysis, the player's own intended **Personal Strategy**, and interactive drills inside a single cohesive loop.

Specifically, seven binding requirements from the brief reshape and extend our initial 50-concept curriculum into a comprehensive **58-Concept Riverline Curriculum Map**:

- **Epistemic Confidence Tiers Replace Binary "Exact vs. Judgement" Labels (Brief §2.1):** Riverline prohibits fake GTO claims, invented EV numbers, and numeric confidence percentages. Every concept and drill output is now mapped to Riverline's four canonical confidence tiers: `Exact (scoped)` (e.g., `Exact · price only` or `Exact · combinatorics`), `Verified` (heuristic benchmarked against solver or combinatorial references within a known tolerance, such as the Rule of 4 & 2), `Estimated` (baseline heuristic or range-dependent judgement), and `Unavailable` (unsolved multiway equilibrium or unobservable live tells).
- **First-User Game Profile Calibration: 30–60 BB Rake-Free Home Games vs. 100 BB+ 9-Max Online (Brief §1):** Standard coaching courses assume 100 BB 6-max raked online games. However, Riverline's primary user plays **6–8 handed home games with 30–60 BB buy-ins and zero rake** alongside **9-max online cash at 100 BB+**. In 30–60 BB cash games, single-raised flop SPR drops from ~13 to ~4–7 and 3-bet flop SPR collapses to 1–2.5, elevating top-pair commitment while crushing speculative implied-odds set-mining. Meanwhile, zero rake makes preflop flatting and multiway pots significantly more profitable. Concepts **#51–#53** are added to model these exact stack and rake regimes.
- **"Unknown is Never Fold, Zero, or Excluded" (Brief §2.2):** When reconstructing live home-game hands or drilling multiway spots, unrevealed opponent hands remain explicitly tagged as `Unknown` rather than silently folded or treated as zero-equity placeholders. Concept **#54** teaches partial-information hand reconstruction.
- **"Heuristic Disagreement is Never a Mistake" & First-Class Personal Strategy (Brief §2.3 & §2.5):** Riverline separates (1) observed play, (2) the player's intended **Personal Strategy**, and (3) the baseline. Drills never scold deviations from an `Estimated` heuristic baseline as "mistakes"; instead, they report descriptive alignment (*inside baseline mix*, *outside baseline mix*, *close*, or *diverges from your Personal Strategy*). Concepts **#55–#56** operationalize this distinction.
- **One Canonical Owner Per Concept in `shared/poker-domain/` (Brief §2.4 & §3):** All mathematical computations live exclusively inside `shared/poker-domain/`; the UI (`app/src/core/logic.js` and ES modules) formats and displays outputs without computing poker math.
- **Trilingual Localization (EN, RU, HE) & Calm Surface Design (Brief §1 & §4):** Drills are engineered for English, Russian, and right-to-left Hebrew (with isolated left-to-right poker notation), following the dark forest / jade / brass palette, one primary action per view, and non-judgmental audio/motion feedback.
- **Legal, Free-First Solver Grounding for M2 → M3 (Brief §2.7, §5 & §6):** We verify the open-source Rust library b-inary/postflop-solver (AGPLv3, offline numerical output generation) and HoldemResources Calculator (HRC Pro) EULA (which explicitly permits commercial distribution of exported calculation results) to transition postflop and preflop spots from `Estimated` to `Verified`.

1. Evidence Table: Mapping Brief Mandates to Curriculum & Product Extensions

In accordance with Section 7 of `RIVERLINE_BRIEF.md` ("Evidence first: open the actual source, quote the relevant clause or line… Mark your own judgement as **[Judgement]**"), the table below links every core mandate from the brief to the exact modifications and additions made in this extended research report.

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Brief Clause	Verbatim Quote from `RIVERLINE_BRIEF.md`	Concrete Extension Added to the Learn Hub Curriculum & Specification
§1 First User & Game Types	"First user: the owner (home games 6–8 handed with 30–60 bb buy-ins and no rake; online 9-max at 100 bb+; studies a lot)."	Added Concepts **#51 (30–60 BB Short/Mid-Stack Cash Geometry)**, **#52 (Rake-Free Home Game Flatting & Multiway Incentives)**, and **#53 (9-Max Full-Ring Early Position Tightness vs. 6–8 Handed)**. Rebuilt Lesson 1, Lesson 6, and Lesson 7 to contrast 40 BB rake-free home games with 100 BB+ raked 9-max online games.
§1 Languages & RTL	"Languages **EN, RU, HE** (Hebrew right-to-left with left-to-right poker notation)."	Added Section 6 specifying bidirectional rendering rules (`<bdi dir="ltr">` around hand combos like `AKs`, board strings, and mathematical ratios) and a trilingual glossary of core poker terms across English, Russian, and Hebrew.
§2.1 Honesty & Confidence Tiers	"Confidence tiers: **Exact** (always scoped, e.g. 'Exact · price only'), **Verified** (heuristic measured against references within a tolerance), **Estimated**, **Unavailable**. No numeric confidence percentages, no fake GTO, no fake EV, no mastery theatre."	Replaced generic "Maths vs. Judgement" labels with a dual classification across all 58 concepts: (1) Cognitive Mode (Exact Maths / Hybrid / Judgement) and (2) Riverline Epistemic Confidence Tier (`Exact · price only`, `Exact · combinatorics`, `Verified · tolerance ±1.5%`, `Estimated · heuristic baseline`, or `Unavailable · unmodeled state`). Added Concept **#57** to teach players these four tiers.
§2.2 Unknown Handling	"**Unknown is never fold, zero or excluded.** Unanswered hands stay unknown; missing information is labelled, never invented."	Added Concept **#54 (Reconstructing Incomplete Hands & Partial-Info Ranges)**. Updated all range and hand-review drills so unrevealed seats remain explicitly tagged as `Unknown` rather than silently folded.
§2.3 & §2.5 Personal Strategy vs. Baseline	"**Heuristic disagreement is never a 'mistake'.** Comparisons are descriptive (inside / outside the baseline's mix, close)… **The player's plan is first-class** (Personal Strategy), separate from what they did and from the baseline."	Added Concept **#55 (Personal Strategy Construction vs. Baseline Mix)** and Concept **#56 (Auditing Observed Play vs. Intended Personal Strategy)**. Replaced "Typical Mistake" scolding in UI drill specs with descriptive 3-way comparison feedback.

Brief Clause	Verbatim Quote from `RIVERLINE_BRIEF.md`	Concrete Extension Added to the Learn Hub Curriculum & Specification
§2.4 & §3 Shared Domain Ownership	"**One canonical owner per concept.** Poker maths live only in the shared domain; the UI formats, never computes... shared poker domain in `shared/poker-domain/`."	Mapped every quantitative concept in the 58-concept table to its canonical module inside `shared/poker-domain/` (e.g., `potOdds.js`, `combinatorics.js`, `spr.js`, `equity.js`) so zero math is duplicated in `app/src/core/logic.js`.
§2.6 Study Tool Boundaries	"**Study tool, not real-time assistance.** A quick 'live decision' mode may exist only with strong warnings; hard limits: no overlays on poker clients..."	Added Concept **#58 (Off-Table Study vs. Ethical Live Decision Boundaries)** and designed post-session Game Night reconstruction workflows rather than in-game RTA prompts.
§4 Design, Motion & Sound	"Look: dark forest / jade / brass, 'calm surface, deep data'... one primary action per view... study cues never sound like right/wrong."	Designed three drill UI archetypes in Section 5 that enforce one primary action per screen, progressive disclosure ("calm surface, deep data"), and neutral acoustic/motion feedback.
§5 Strategy Engine & Solvers	"**b-inary/postflop-solver** (AGPL) run offline, shipping only numerical outputs... **HRC Pro** allows commercial use of exported results... Gap: preflop and multiway for the owner's game types."	Verified the exact licensing clauses of b-inary/postflop-solver and HoldemResources Calculator EULA in Section 6, and proposed an offline precomputation pipeline for 30–60 BB no-rake and 100 BB 9-max spots.

2. Pedagogical Sources & Literature Grounding

Every concept definition, strategic leak description, and practice exercise in this report is written in original prose synthesized from authoritative poker literature and coaching curricula. No proprietary lesson text, chart matrices, or quiz items have been copied:

- **Foundational Mathematical Theory & Expected Value:** Synthesized from David Sklansky's *The Theory of Poker*, which established the Fundamental Theorem of Poker, expected value (EV), direct pot odds, implied odds, and semi-bluffing math.
- **Effective Stacks, SPR & Commitment Geometry:** Adapted from Matt Flynn, Sunny Mehta, and Ed Miller's *Professional No-Limit Hold'em, Volume I*, which formalized Stack-to-Pot Ratio (SPR), effective stack ceilings, and preflop bet-sizing geometry to manipulate flop SPR.
- **Online Cash Game Range Construction & Postflop Reasons to Bet:** Informed by Peter Clarke's *The Grinder's Manual* and the Run It Once *From The Ground Up* cash curriculum, supplemented by track structures from the Upswing Poker Lab, Red Chip Poker *CORE 2.0*, and Jonathan Little's PokerCoaching.com Cash Game Masterclass.
- **Game Theory Optimal (GTO) & Solver Heuristics:** Grounded in Matthew Janda's *Applications of No-Limit Hold'em*, Michael Acevedo's *Modern Poker Theory*, and Andrew Brokos's *Play Optimal Poker*.
- **Live/Home Game Exploits, Physical Tells & Session Mindset:** Drawn from Ed Miller's *The Course*, Alexander Fitzgerald's *Exploitative Play in Live Poker*, Zachary Elwood's *Reading Poker Tells*, and Jared Tendler's *The Mental Game of Poker*.

3. Calibration for Riverline's Dual Game Environments (30–60 BB Home vs. 100 BB+ 9-Max Online)

A critical insight from Section 1 of `RIVERLINE_BRIEF.md` is that the owner's two primary cash environments sit at opposite ends of the structural spectrum. If Riverline's Learn Hub teaches generic 100 BB 6-max raked heuristics as universal truth, it will actively mislead players in both of the owner's actual games. The table below contrasts the mathematical and strategic realities of both environments:

Structural Dimension	Home Game Profile: 6–8 Handed, 30–60 BB Buy-Ins, Zero Rake	Online Cash Profile: 9-Max Full Ring, 100 BB+ Stacks, Standard Rake
Single-Raised Pot (SRP) Flop SPR	**Low-to-Medium SPR (3.5 to 7.5):** A 3 BB open called by one player at 40 BB effective creates a 7.5 BB pot with 37 BB behind (`SPR = 4.9`). If two players call, the pot is 10.5 BB with 37 BB behind (`SPR = 3.5`). Top pair with a strong kicker is routinely a 2-street stack-off hand.	**High SPR (11 to 14+):** A 2.5 BB open called by the Big Blind at 100 BB effective creates a 5.5 BB pot (less rake) with 97.5 BB behind (`SPR = 17.7`). Stacking off with one pair in a single-raised pot is a severe leak.
3-Bet Pot Flop SPR	**Micro SPR (0.8 to 2.2):** A 3 BB open 3-bet to 9 BB at 40 BB effective leaves 31 BB behind into an 18.5 BB pot (`SPR = 1.67`). Any overpair or top pair (`AK`, `AQ`) is mathematically committed on the flop.	**Low-to-Medium SPR (3.5 to 4.8):** A 2.5 BB open 3-bet to 8 BB called at 100 BB leaves 92 BB into a 16.5 BB pot (`SPR = 5.5`), still requiring turn/river barrelling discipline.
Set-Mining & Speculative Implied Odds	**Severely Devalued:** With 35–50 BB stacks, small pocket pairs (`22`–`66`) and low suited connectors (`54s`, `65s`) lack the 15x–20x stack depth needed to set-mine profitably against 3–4 BB raises. High-card equity (`KQo`, `AJo`, `ATs`) dominates.	**High Value in Late Position:** At 100–150 BB stacks, small pocket pairs and suited connectors profitably call early-position opens in position or multiway to target overpairs.
Preflop Flatting & Rake Impact	**Zero Rake Expands Flatting:** Because 0% is dropped from the pot on the flop, cold-calling in position and defending the Big Blind or Small Blind gain ~0.5 to 1.5 BB/100 in pure EV compared to raked games.	**Rake Penalizes Flatting:** A 5% rake (capped at 3–4 BB) taxed only when a flop is seen ("no flop, no drop") forces tight 3-bet-or-fold strategies from MP, CO, BTN, and SB.
Positional & Multiway Frequency	**High Multiway Frequency:** Casual home-game players limp and cold-call frequently, creating 3-way to 5-way flops where pure bluffs fail and protection/value sizing dominates.	**Tight Early Seats (UTG, UTG+1, UTG+2):** Having 8 opponents left to act requires a 9%–12% UTG opening range; pots are predominantly heads-up between the open-raiser and one caller or 3-bettor.

4. Complete 58-Concept Curriculum Map (Beginner → Advanced)

To keep the 58-concept map effortless to read and scan across print pages, the progression is organized into six modular stage tables: **Tier 1 (Foundations, Position, Pot Odds & Equity, #1–#10)**, **Tier 2 (Ranges, Preflop Architecture & SPR, #11–#20)**, **Tier 3 (Postflop Value, Bluffing & Range Morphology, #21–#34)**, **Tier 4 (Blockers, Indifference & Sizing Geometry, #35–#43)**, **Tier 5 (Exploitative Play, Tells & Session Reading, #44–#50)**, and **Tier 6 (Riverline Workstation, 30–60 BB Home Game & Epistemic Literacy Extensions, #51–#58)**. Every concept includes its prerequisites, original one-line definition, typical mistake (framed descriptively for drills), original exercise idea, cognitive mode, Riverline Confidence Tier, and primary literature citation.

#	Concept & Prereqs	One-Line Definition (Original)	Typical Leak / Baseline Divergence	Quick Exercise Idea (Original)	Maths vs. Judgement & Riverline Tier
1	**Relative vs. Absolute Hand Strength** *Prereq:* NLHE Rules	Evaluating a hand's strength relative to board texture and multi-street action rather than its raw rank on a chart (*Theory of Poker*).	Overplaying a bare overpair or two pair on a four-flush or four-straight board facing heavy aggression.	Rank five pocket pairs (AA–66) from strongest to weakest across three river runouts (K-Q-J-T-2, 7-7-2-2-3, A-9-8-7-6 suited).	**Judgement** Tier: `Estimated · board interaction`
2	**Table Positions (Absolute vs. Relative)** *Prereq:* None	Acting later in the betting order to gather information before committing chips, distinguishing button seat from acting last relative to the aggressor (*Grinder's Manual*).	Calling preflop from the Small Blind or early seats with dominated offsuit broadways because the cards look appealing.	Map a 6-max, 8-handed home, and 9-max table diagram; trace how acting order shifts when the preflop aggressor is on the Cutoff vs. Small Blind.	**Judgement** Tier: `Exact · seat order` / `Estimated · EV impact`
3	**Effective Stack Size** *Prereq:* #2	The smaller stack between two active players in a hand, which strictly caps the maximum chips that can be won or lost (*Professional NLHE*).	Making a speculative set-mine call against a 35 BB home-game stack while looking only at your own 150 BB stack.	Given four stacks at a home game (35 BB, 55 BB, 110 BB, 180 BB), compute the exact effective stack for every heads-up and 3-way pairing.	**Exact Maths** Tier: `Exact · stack cap` (`shared/poker-domain`)
4	**Expected Value (EV) & Long-Run Thinking** *Prereq:* #3	The probability-weighted average payoff of an action over infinite trials, separating decision quality from single-hand variance (*Theory of Poker*).	Judging a play by its immediate showdown outcome ("resulting") after an opponent hits a 4-outer on the river.	Compute the exact dollar EV of a $40 river call into a $120 pot when your hand wins 30% of the time and loses 70% of the time.	**Exact Maths** Tier: `Exact · given equity` (`EV = ∑ P·Payoff`)
5	**Pot Odds & Breakeven Required Equity** *Prereq:* #4	The ratio of the call cost to the final pot after calling, expressed as the minimum win percentage required for a 0-EV call (Red Chip *CORE 2.0*).	Confusing ratio notation (3:1) with percentage (thinking 3:1 means 33% instead of 25%) or omitting your call from the denominator.	Convert five bet sizes (25%, 33%, 50%, 75%, 100% pot) into exact breakeven equity percentages in under 60 seconds.	**Exact Maths** Tier: `Exact · price only` (`Call / (Pot+Bet+Call)`)
6	**Outs & Rule of 4 and 2 (Raw Draw Equity)** *Prereq:* #5	Counting unseen cards that improve your hand to the winner and multiplying by 4 on the flop (or 2 on the turn) to estimate draw equity (*Theory of Poker*).	Multiplying outs by 4 on the flop when facing a bet where the opponent will fire again on the turn, denying a free river card.	Deal 10 flop hands with flush, straight, or combo draws and compare your Rule of 4/2 mental estimate against exact combinatorics.	**Exact Maths** Tier: `Exact · combinatorics` vs. `Verified · Rule 4/2 (±2%)`

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7	**Dirty Outs & Redraw Discounting** *Prereq:* #6	Subtracting or weighting down outs that simultaneously complete a superior hand for the opponent, such as pairing the board against a set (PokerCoaching Masterclass).	Counting all 9 flush outs and 6 overcard outs as 15 clean outs on a paired or monotone-threatened board.	Audit five flop draw scenarios and discount raw out counts by assigning weights (0, 0.5, or 1.0) based on the opponent's value range.	**Hybrid** Tier: `Estimated · out weights` + `Exact · adjusted %`
8	**Implied Odds & Reverse Implied Odds** *Prereq:* #3, #5, #7	Adjusting direct pot odds for extra chips won on later streets when hitting (implied) vs. chips lost when making a second-best hand (*Professional NLHE*).	Calling preflop raises with dominated offsuit broadways (`KJo`, `ATo`) citing "implied odds" when they suffer from reverse implied odds.	Calculate the exact minimum turn/river payoff needed to justify calling a 3 BB raise with `44` when direct pot odds fall 8% short.	**Hybrid** Tier: `Exact · required payoff` + `Estimated · villain payoff`
9	**Hand Ranges & the 13×13 Matrix** *Prereq:* #1, #2	Representing a player's holdings as a spectrum of hand combinations on a 169-cell grid rather than guessing one two-card hand (RIO *From The Ground Up*).	Putting an opponent on a single exact hand ("he has `AK`") and making severe river errors when he shows up with a set.	Shade a blank 13×13 grid to build a 15% early-position opening range, separating suited (above diagonal), pairs (diagonal), and offsuit (below).	**Hybrid** Tier: `Exact · grid combos` + `Estimated · villain range`
10	**Combinatorics (1,326 Combos & Card Removal)** *Prereq:* #9	Counting exact ways to deal hands (6 per pair, 4 per suited hand, 12 per offsuit hand) and reducing counts for visible dead cards (*Modern Poker Theory*).	Treating `AA` and `AKo` as equally likely inside an opponent's 3-bet range despite `AKo` having twice as many combinations.	Given a flop of `As-Kh-7d` and your hand `Ah-Qd`, compute the exact remaining combos of `AA`, `KK`, `77`, `AKs`, and `AKo`.	**Exact Maths** Tier: `Exact · combinatorics` (`shared/poker-domain`)
11	**Hand vs. Range Equity** *Prereq:* #6, #9, #10	Computing a specific hand's combo-weighted win percentage against every hand combination inside an opponent's assigned range (*Applications of NLHE*).	Evaluating top pair only against the opponent's bluffs while ignoring how many value combinations dominate you.	Take `Ah-Jh` on `Jd-8h-4c` and compute weighted equity against a 3-hand toy range: `AA` (6 combos), `KJs` (3 combos), and `Th9h` (1 combo).	**Exact Maths** Tier: `Exact · given range` (`shared/poker-domain`)

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12	**Reasons to Bet: Value vs. Bluff** *Prereq:* #4, #11	Betting strictly to get called by worse hands (value) or to fold out better hands (bluff), with equity denial as a secondary benefit (*Grinder's Manual*).	Betting a medium-strength hand "to see where I'm at," which folds out worse hands and gets called only by better ones.	Review 10 flop and turn spots with middle pair or weak top pair; list which worse hands call and which better hands fold if you bet.	**Judgement** Tier: `Estimated · strategic intent`
13	**Preflop Open-Raising (RFI) by Position** *Prereq:* #2, #9	Constructing disciplined, position-graded Raise-First-In ranges that widen progressively from early position to the Button (Upswing Poker Lab).	Open-limping in early or middle position to "see a cheap flop," inviting multiway squeeze pots out of position.	Build baseline RFI charts for 6-max and 9-max (~11% UTG 9-max vs. ~16% UTG 6-max, ~45% BTN) and drill 20 borderline hands by seat.	**Hybrid** Tier: `Estimated · heuristic baseline` (M2) → `Verified` (M3)
14	**Isolating Limpers (Home & Loose Online)** *Prereq:* #2, #12, #13	Raising over weak open-limpers with a wider value range in position to play a heads-up pot with initiative against a capped player (*The Course*).	Over-limping behind three recreational limpers with `AQo` or `JJ`, letting five players realize free equity on the flop.	Size isolation raises using the standard heuristic (`3 BB + 1 BB/limper + 1 BB OOP` online; `4–5 BB + 1–2 BB/limper` live).	**Hybrid** Tier: `Estimated · sizing heuristic`
15	**3-Betting: Linear vs. Polarized Ranges** *Prereq:* #10, #13	Re-raising preflop with either a merged top-of-range spectrum (linear, vs. loose callers) or premiums plus blocker bluffs (polarized, vs. folders/4-bettors) (*Modern Poker Theory*).	3-betting low suited connectors against a loose home-game caller who never folds to 3-bets, instead of linear value like `KQs` and `AJo`.	Construct two Cutoff vs. MP 3-bet grids: one linear range against a recreational caller and one polarized range against a reg who folds 60%.	**Judgement** Tier: `Estimated · range architecture`
16	**Big Blind Defense & Closing Action** *Prereq:* #5, #8, #13	Defending wide from the Big Blind against single raises because 1 BB is already invested and you close the preflop action heads-up (RIO *From The Ground Up*).	Overfolding the Big Blind to a 2.2x–2.5x Button open out of fear of playing OOP, surrendering automatic pot-odds profitability.	Compute exact pot odds facing a 2.5 BB Button open from the Big Blind (0% rake home game vs. 5% raked online) and compare defense thresholds.	**Exact Maths** Tier: `Exact · price only` + `Estimated · R factor`

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17	**Rake & Pre-Flop Structure Adjustments** *Prereq:* #4, #13, #16	Tightening preflop flat-calling ranges in high-rake online/casino games while expanding flatting ranges in rake-free home games (*Grinder's Manual*).	Copying a 3-bet-or-fold high-rake online chart into a rake-free home game (or flatting wide in a 10% micro-stakes rake trap).	Calculate the exact effective pot odds reduction when a 5% rake (capped at 3 BB) is removed from a 6.5 BB flop pot vs. a 0% rake pot.	**Exact Maths** Tier: `Exact · net pot after drop`
18	**Stack-to-Pot Ratio (SPR) & Commitment** *Prereq:* #3, #8	Dividing the effective stack on the flop by the flop pot size to determine whether one-pair hands should stack off or pot-control (*Professional NLHE*).	Stacking off 100 BB in a single-raised pot (SPR ~13) with unimproved `AA` or top pair on a coordinated board.	Categorize 6 scenarios into Low SPR (1–3), Medium SPR (4–7, typical 40 BB home SRP), and High SPR (10+), defining your commitment floor.	**Exact Maths** Tier: `Exact · ratio (Effective_Stack / Flop_Pot)`
19	**Equity Realization (R) & Playability** *Prereq:* #2, #11, #18	The ratio of actual EV captured to raw all-in equity, governed by position, suitedness, connectivity, SPR, and skill edge (*Play Optimal Poker*).	Calling out of position with raw-equity offsuit hands (`K7o`) that have 45% raw equity against a Button open but realize <60% of it.	Rank four hands with ~42% raw equity against a tight range (`A2o`, `K9o`, `76s`, `55`) by expected equity realization IP vs. OOP.	**Hybrid** Tier: `Estimated · realization factor R`
20	**Flop Texture Classification (Static vs. Dynamic)** *Prereq:* #6, #11	Grouping flops by connectivity, suitedness, and high-card rank into static (`K-7-2r`, stable nuts) vs. dynamic (`J-T-8ss`, shifting nuts) (Upswing Poker Lab).	Slowplaying vulnerable top two pair on a dynamic `T-9-7` two-tone board where half the deck changes the nuts on the turn.	Classify 12 flops along two axes (Static vs. Dynamic and Dry vs. Wet) and count how many turn cards alter the nut hand.	**Judgement** Tier: `Estimated · board taxonomy`
21	**Range Advantage vs. Nut Advantage** *Prereq:* #11, #20	Distinguishing higher average equity across the whole range (dictates bet frequency) from exclusive nut combo density (dictates bet size) (*Modern Poker Theory*).	Firing a 100% pot c-bet on `K-7-2r` (where you hold range advantage, not exclusive nut advantage) or betting 25% on `T-9-6ss`.	Compare Button RFI vs. BB Caller on `A-K-4r`, `8-7-5ss`, and `2-2-7r`; identify who holds Range Advantage and Nut Advantage on each.	**Hybrid** Tier: `Exact · given ranges` / `Estimated · heuristic`

#	Concept & Prereqs	One-Line Definition (Original)	Typical Leak / Baseline Divergence	Quick Exercise Idea (Original)	Maths vs. Judgement & Riverline Tier
22	**Continuation Betting (IP vs. OOP)** *Prereq:* #12, #20, #21	Following up preflop aggression with high-frequency small bets on static high-card boards IP, while checking heavily OOP on low connected flops (RIO *From The Ground Up*).	Auto-c-betting 75% pot on `7-6-4` two-tone after opening UTG against a Big Blind caller who holds the nut advantage.	Assign five UTG vs. BB single-raised flops to either High-Frequency Small C-Bet (33%), Polarized Big C-Bet (66%+), or Range Check.	**Judgement** Tier: `Estimated · baseline mix` (M2) → `Verified` (M3)
23	**Fold Equity & Bluff Breakeven Math** *Prereq:* #4, #5, #12	The probability an opponent folds to a bet, compared against the exact risk-to-reward breakeven percentage of a zero-equity bluff (*Applications of NLHE*).	Firing a half-pot river bluff against a calling station who folds <20% of the time (when half-pot requires 33.3% folds to break even).	Use `Risk / (Risk + Reward)` to calculate required fold percentages for 33%, 50%, 75%, 100%, and 150% pot bluffs.	**Exact Maths** Tier: `Exact · price only` (`Bet / (Pot + Bet)`)
24	**Semi-Bluffing & Total Bluff EV** *Prereq:* #6, #19, #23	Betting or raising with a draw that combines immediate fold equity with draw equity when called (*Theory of Poker*).	Semi-bluff jamming against a short-stacked opponent with zero fold equity (SPR 1.2 with an overpair) when calling has higher EV.	Compute total EV using `P(Fold)·Pot + P(Call)·[Eq·NewPot - Cost]` for a 9-out flush draw check-raise jam on the flop.	**Exact Maths** Tier: `Exact · given fold % & equity`
25	**Backdoor Equity & Float Planning** *Prereq:* #6, #22, #24	Valuing hands with overcards plus 3-flush and 3-straight potential on the flop, which turn into strong barrels on ~10–12 turn cards (*Grinder's Manual*).	Floating a flop c-bet with zero backdoor outs (`K8o` no suit on `J-9-4ss`) and check-folding every turn card.	Count the exact number of turn cards (out of 47) that give `Ks-Qs` top pair, a flush draw, or an open-ended straight draw on `Jd-9s-3h`.	**Exact Maths** Tier: `Exact · turn out count`
26	**Thin Value Betting & Fat Value** *Prereq:* #11, #12	Betting on the river when your hand beats more than 50% of the opponent's calling range, even with second pair or weak top pair (*Exploitative Play*).	Checking back the river in position with `A-J` on `J-8-4-2-6` out of fear of a slowplayed set, missing value from worse jacks.	List all combos in a BB caller's river range, cross out combos that fold to a 50% bet, and verify whether you beat >50% of callers.	**Hybrid** Tier: `Exact · >50% threshold` + `Estimated · call range`

#	Concept & Prereqs	One-Line Definition (Original)	Typical Leak / Baseline Divergence	Quick Exercise Idea (Original)	Maths vs. Judgement & Riverline Tier
27	**Equity Denial & Protection Betting** *Prereq:* #11, #12, #19	Betting vulnerable made hands (88 on 7-5-2) to fold out overcard hands (QJo, KTo) holding ~24% raw equity against you (*Applications of NLHE*).	Checking back vulnerable mid-pairs in multiway home-game pots and letting three opponents realize 50%+ combined equity for free.	Calculate the exact EV gained when forcing an opponent on KQo (24% equity) to fold on 9-6-3 into a $40 pot when holding 77.	**Exact Maths** Tier: `Exact · denied equity EV` (`Eq_folded × Pot`)
28	**Pot Control & Showdown Value (SDV)** *Prereq:* #12, #18, #26	Checking medium-strength hands (marginal top pair, second pair) to keep the pot manageable, induce bluffs, and reach showdown (Red Chip *CORE 2.0*).	Turning a showdown-value hand like Q-J on Q-8-3 into a 3-street barrel that folds out worse hands and isolates yourself vs. sets.	Sort 12 flop hands on K-9-4r into four buckets: Clear Value Bet, Clear Check-Back (SDV), Semi-Bluff Bet, and Give Up.	**Judgement** Tier: `Estimated · bucket classification`
29	**Condensed vs. Polarized Ranges** *Prereq:* #15, #26, #28	Recognizing when a range consists of medium bluff-catchers (condensed, after checking/calling) vs. nut hands and air (polarized, after big betting) (*Play Optimal Poker*).	Firing a 25% river bet with a polarized nut-or-nothing range, giving a condensed bluff-catching range cheap odds to call.	Identify which player is polarized and which is condensed after a Flop Bet-Call, Turn Bet-Call, River Check sequence on a brick runout.	**Judgement** Tier: `Estimated · range morphology`
30	**Capped vs. Uncapped Ranges** *Prereq:* #21, #29	Identifying when a player's passive line (flatting preflop or checking back two streets) removes all nut combinations from their range (Upswing Poker Lab).	Calling down three big barrels with one pair when the opponent's range is uncapped and your own range is capped at top pair.	Trace a hand from preflop to river and pinpoint the exact street and action where one player's range becomes capped at one pair.	**Judgement** Tier: `Estimated · deductive cap check`
31	**Turn Barreling & Dynamic Runout Shifts** *Prereq:* #20, #22, #25	Selecting turn cards to fire a second barrel based on whether the card favors the raiser's range (overcards), adds draw equity, or bricks (RIO *From The Ground Up*).	Firing a mechanical second barrel on a low card that pairs the board (7-6-3-6) after the Big Blind called on the flop.	Group all 49 unseen turn cards after a J-7-2r flop into Scare/Range Cards, Equity-Shift Cards, Board-Pairing Cards, and Bricks.	**Judgement** Tier: `Estimated · turn runout shift`

#	Concept & Prereqs	One-Line Definition (Original)	Typical Leak / Baseline Divergence	Quick Exercise Idea (Original)	Maths vs. Judgement & Riverline Tier
32	**Check-Raising (Flop & Turn Construction)** *Prereq:* #18, #21, #24	Balancing strong value hands (sets, two pair) with high-equity semi-bluffs and natural blocker bluffs when defending OOP (*Modern Poker Theory*).	Only check-raising the flop with sets and two pair, making your check-raise transparent so regs fold top pair immediately.	Construct a balanced Big Blind check-raise portfolio on `T-8-4` two-tone using roughly a 2:1 bluff-to-value combo ratio on the flop.	**Hybrid** Tier: `Exact · combo ratio` + `Estimated · draw selection`
33	**Delayed C-Betting & Probe Betting** *Prereq:* #22, #28, #30	Betting the turn IP after checking back the flop (delayed c-bet) or leading the turn OOP after the preflop raiser checks back (probe bet) (PokerCoaching Masterclass).	Treating a flop check-back as an automatic surrender of the pot, missing profitable turn stabs after the opponent checks twice.	Design a turn probe-betting mix from the Big Blind after the Button checks back a `Q-7-3` flop and the turn brings a `5`.	**Judgement** Tier: `Estimated · heuristic baseline`
34	**Multiway Pot Mechanics (Home & Live Staple)** *Prereq:* #2, #19, #27	Tightening c-bet frequencies, sizing down or checking without nut potential, and respecting compounded defensive burden in 3+ player pots (*The Course*).	Bluff c-betting a dry `A-8-3` flop with `KQo` into three home-game callers, forgetting someone almost always holds an Ace multiway.	Compute the compound probability that at least one of three opponents holds a top-20% hand (`1 - 0.8^3 = 48.8%`).	**Hybrid** Tier: `Exact · compound probability` / `Unavailable · multiway GTO`
35	**Blockers (Blocking Value vs. Unblocking Bluffs)** *Prereq:* #10, #23, #29	Using hole cards to remove nut combos from villain's calling range (when bluffing) or preserve missed draws (when bluff-catching) (*Modern Poker Theory*).	Calling a river shove with `Ah-Jh` on a missed heart flush board, where holding `Ah` and `Jh` removes the villain's missed heart bluffs.	On `Ks-Ts-4s-8d-2c`, compare bluffing with `As-9d` (nut spade blocker) vs. `9s-8s` or `Qh-Jd` and count removed villain flush combos.	**Exact Maths** Tier: `Exact · combinatorics` (`shared/poker-domain`)
36	**Bluff-to-Value Ratios & River Indifference** *Prereq:* #5, #23, #29	Balancing a polarized river bet so the bluff proportion matches the pot odds offered to villain's bluff-catchers (`α = Bet / (Pot + 2·Bet)`) (*Applications of NLHE*).	Reaching the river with 18 missed draw combos and betting all 18 alongside only 6 value combos, overbluffing massively.	Given 12 nut value combos on the river, compute the exact number of bluff combos needed for a 50% bet, 75% bet, and 150% overbet.	**Exact Maths** Tier: `Exact · toy game indifference`

#	Concept & Prereqs	One-Line Definition (Original)	Typical Leak / Baseline Divergence	Quick Exercise Idea (Original)	Maths vs. Judgement & Riverline Tier
37	**Minimum Defense Frequency (MDF) & Limits** *Prereq:* #23, #36	Calculating the range fraction required to prevent 0-equity bluff profit (`Pot / (Pot + Bet)`), while knowing when range asymmetry overrides MDF (*Play Optimal Poker*).	Defending 67% of your range on the flop against a half-pot c-bet when the board heavily favors the aggressor, bleeding EV.	Compute MDF for 33%, 50%, 75%, and 100% pot bets, then compare against solver over-folding on an `A-K-Q` flop for the Big Blind.	**Hybrid** Tier: `Exact · MDF formula` + `Verified · solver exception`
38	**Geometric Bet Sizing & Stack-Off Math** *Prereq:* #18, #21, #29	Choosing a uniform pot-percentage bet across n remaining streets so the entire effective stack goes all-in smoothly on the river (*Modern Poker Theory*).	Betting 33% pot on the flop and turn with a set at 150 BB deep, leaving an awkward 2.5x pot jam on the river to win the stack.	Given a $20 flop pot and $180 effective stacks (SPR 9), solve `(1 + 2b)^3 = 1 + 2(SPR)` to find the exact uniform bet size (~62% pot).	**Exact Maths** Tier: `Exact · geometric formula` (`shared/poker-domain`)
39	**Overbetting (Polarized Nut Advantage)** *Prereq:* #21, #30, #36, #38	Betting >100% pot on the turn or river when holding an exclusive monopoly on the nuts against a capped opponent range (Upswing Poker Lab).	Overbetting 150% pot on a board where villain can hold the nuts, or overbetting with a merged one-pair hand.	Identify three turn/river runouts where the BB caller is capped at two pair while the raiser holds top set/straights, and build a 125% range.	**Hybrid** Tier: `Exact · bluff ratio` + `Estimated · cap diagnosis`
40	**Block Betting (Small OOP River Sizing)** *Prereq:* #26, #28, #35	Betting 20%–30% pot OOP on the river with medium hands (plus nut traps and bluffs) to get thin value and avoid facing a large IP bet (*Modern Poker Theory*).	Always checking OOP with second pair or weak top pair, letting IP check back worse hands or blast you off your equity.	Build a 25% pot river block-bet split from the Big Blind on a brick river, mixing thin value, nut check-raise traps, and blocker bluffs.	**Judgement** Tier: `Estimated · heuristic baseline` (M2) → `Verified` (M3)
41	**River Bluff-Catching (Theory vs. Pool)** *Prereq:* #29, #35, #36, #37	Combining blocker removal with population tendencies to decide whether to call or fold a condensed bluff-catcher on the river (*Exploitative Play*).	Calling a recreational player's river check-raise with top pair citing "MDF and blockers," ignoring that live river raises are <5% bluffs.	Evaluate three bluff-catchers (`As-Kd`, `Kh-Qd`, `Kd-Jd`) on `Kc-9s-8s-3d-2h` under equilibrium vs. an under-bluffing pool.	**Hybrid** Tier: `Exact · blocker count` + `Estimated · pool bluff %`

#	Concept & Prereqs	One-Line Definition (Original)	Typical Leak / Baseline Divergence	Quick Exercise Idea (Original)	Maths vs. Judgement & Riverline Tier
42	**4-Betting, 5-Betting & Low-SPR 3-Bet Pots** *Prereq:* #15, #18, #35	Constructing polarized 4-bet ranges using high-card blockers (A5s, KQo) alongside value (AA–QQ, AK) and navigating SPR 2–4 postflop (RIO *From The Ground Up*).	Flatting 3-bets OOP with dominated offsuit broadways (AQo, KJo) or 4-betting medium pairs (88, 99) that play better as calls.	Build a Button vs. SB 4-bet range at 100 BB using AA-QQ, AK for value and select exact blocker bluff combos.	**Hybrid** Tier: Exact · combo counts + Verified · HRC Pro preflop
43	**Deep-Stack Cash Dynamics (200 BB+ & Straddles)** *Prereq:* #8, #18, #38	Shifting weight toward nut-potential hands (suited Aces, suited connectors, pocket pairs) while devaluing one-pair hands as SPR doubles (PokerCoaching Masterclass).	Playing TPTK (AK on K-8-4) for 250 BB stacks the exact same way as at 100 BB, getting stacked by sets and wraps.	Recalculate effective stack depth when a $1/$2 ($500 stacks) game puts on a $5 button straddle (250 BB → 100 effective straddles).	**Hybrid** Tier: Exact · effective straddles + Estimated · nut floor
44	**Player Profiling & Archetype Exploitation** *Prereq:* #12, #26, #41	Categorizing opponents into actionable archetypes (Nit, Calling Station, Loose-Passive, Maniac, TAG/LAG) and applying targeted deviations (*The Course*).	Running a balanced triple-barrel bluff strategy against a loose-passive home-game station whose river fold rate is 15%.	Create a 2×2 grid (VPIP: Loose vs. Tight; Aggression: Passive vs. Aggressive) and write the #1 high-EV adjustment for each quadrant.	**Judgement** Tier: Estimated · archetype exploit
45	**Node Locking & Exploiting Pool Leaks** *Prereq:* #36, #41, #44	Shifting from mixed equilibrium frequencies to pure max-EV strategies (100% bet or 100% fold) once an opponent's threshold leak is known (*Play Optimal Poker*).	Mixing 50% call / 50% fold on the river with a bluff-catcher when you know the pool under-bluffs that line by 12% (where pure fold is max-EV).	Take a river spot where a 75% bet requires 30% bluffs; compute your EV shift if villain bluffs 18% and switch to 100% fold.	**Hybrid** Tier: Exact · given node lock (b-inary/postflop-solver)
46	**Live Physical, Timing & Bet-Sizing Tells** *Prereq:* #29, #44	Correlating non-verbal cues, snap-checks, tank-calls, and non-standard bet sizes with hand strength in live and online games (*Reading Poker Tells*).	Relying on an isolated Hollywood tell without establishing a baseline or checking the player's logical range on the board.	Log three reliable patterns: instant flop check-backs (capped medium hands), micro-sizing by recreationals, and posture relaxation.	**Judgement** Tier: Unavailable · unmodeled behavioral cue

#	Concept & Prereqs	One-Line Definition (Original)	Typical Leak / Baseline Divergence	Quick Exercise Idea (Original)	Maths vs. Judgement & Riverline Tier
47	**Table Selection, Seat Selection & Image** *Prereq:* #2, #3, #44	Maximizing win rate by choosing games with high-VPIP players, sitting directly left of the deepest whale, and tracking your own table image (*Exploitative Play*).	Staying out of ego at a reg-heavy table or sitting directly to the right of a deep-stacked maniac who acts after you every hand.	Draw an 8-handed table with two loose whales (seats 3 and 7) and three tight regs; rank the open seats from highest EV to lowest EV.	**Judgement** Tier: `Estimated · positional meta-strategy`
48	**Reading a Session: Momentum, Tilt & Stop-Loss** *Prereq:* #4, #44, #47	Auditing table dynamics (stack shifts, winners locking up, losers tilting) and your own cognitive state (A-game vs. C-game, fatigue) in real time (*Mental Game of Poker*).	Chasing losses ("getting unstuck") late at night after the recreational players leave and your focus drops to C-game autopilot.	Run a 5-point mid-session audit every 60 minutes: profit source, positional seat, ranged decisions vs. guessing, tilt triggers, stay/quit.	**Judgement** Tier: `Unavailable · metacognitive state`
49	**Variance, Win-Rate (BB/100) & Bankroll Math** *Prereq:* #3, #4, #17	Sizing your poker bankroll (20–30 buy-ins for soft live/home, 40–60+ for online) to bound Risk of Ruin given win rate and standard deviation (*Modern Poker Theory*).	Moving up in stakes after a 1,000-hand heater, mistaking short-term standard deviation for a sustainable 15 BB/100 true win rate.	Using standard deviation `σ = 85 BB/100` and win rate `5 BB/100`, compute the 95% confidence interval (±2σ) over 10,000 hands.	**Exact Maths** Tier: `Exact · statistical formula (shared/poker-domain)`
50	**Off-Table Hand Review & Database Audit** *Prereq:* #11, #36, #45	Structuring post-session study by tagging ambiguous hands, reconstructing ranges, and comparing lines against Personal Strategy and baselines (*Grinder's Manual*).	Only reviewing big all-in coolers (`KK` vs. `AA`) instead of auditing medium turn and river spots where recurring EV leaks occur.	Execute a 4-step hand review: (1) Assign villain range, (2) Count value vs. bluff combos, (3) Check pot odds/blockers, (4) Compare to Personal Strategy.	**Hybrid** Tier: `Exact · math checks` + `Estimated · baseline mix`
51	**30–60 BB Short/Mid-Stack Cash Geometry** *Prereq:* #3, #8, #18	Adapting preflop and flop play to 30–60 BB buy-ins where single-raised flop SPR is 4–7 and 3-bet flop SPR is 1–2.5, boosting high-card top-pair equity over speculative set-mining (*Professional NLHE*; Brief §1).	Calling a 3.5 BB open at 40 BB effective with `33` or `65s` hoping to "win a big pot," when the effective stack is too shallow to pay off a 7.5-to-1 set draw.	Compare the EV of `KQo` vs. `44` facing a 3 BB open at 40 BB effective vs. 120 BB effective, computing exact flop SPR and required implied payoff in each.	**Exact Maths** Tier: `Exact · SPR & implied shortfall` + `Verified · HRC Pro`

#	Concept & Prereqs	One-Line Definition (Original)	Typical Leak / Baseline Divergence	Quick Exercise Idea (Original)	Maths vs. Judgement & Riverline Tier
52	**Rake-Free Home Game Flatting & Multiway Shifts** *Prereq:* #14, #16, #17, #34	Widening preflop cold-calling, Big Blind defense, and Small Blind completion ranges when playing in a 0% rake home game compared to high-rake online pools (*Modern Poker Theory*; Brief §1).	Using a rigid 3-bet-or-fold raked online chart in a rake-free home game, folding profitable suited broadways and pocket pairs that realize full pot equity with zero house drop.	Toggle Riverline's Table Rules between "5% Rake (3 BB Cap)" and "0% Rake (Home Game)" on a Button vs. Cutoff 2.5 BB open and identify the 6–8 extra combos that enter the calling mix.	**Hybrid** Tier: `Exact · 0% rake price` + `Estimated · heuristic baseline`
53	**9-Max Full-Ring Early Position Discipline vs. 6–8 Handed** *Prereq:* #2, #10, #13	Calibrating early-position opening and 3-bet defense ranges for 9-max tables (where UTG faces 8 uncapped opponents) versus 6–8 handed tables (PokerCoaching Masterclass; Brief §1).	Opening `KJo`, `ATo`, or `22` from UTG at a 9-max table, running into a ~32% combined probability that one of the 8 remaining players wakes up with a top-5% hand.	Compute the probability that at least one of `N` remaining players holds a top-5% hand for `N = 8` (9-max UTG: `33.7%`) vs. `N = 5` (6-max UTG: `22.6%`).	**Exact Maths** Tier: `Exact · compound probability` (`1 - 0.95^N`)
54	**Reconstructing Incomplete Hands & Unknown Preservation** *Prereq:* #9, #10, #50	Rebuilding live or home-game hands where unrevealed opponent cards and unrecorded stack depths stay explicitly tagged as `Unknown` rather than invented or folded (Brief §2.2).	Forcing a guessed hand onto an opponent who mucked at showdown or treating an unremembered preflop limper as folded when computing pot size and equity bounds.	Reconstruct a 4-way home-game hand in Riverline where one caller's hole cards are known, one mucked on the river, and one folded the turn; compute exact pot odds and bounded equity intervals.	**Hybrid** Tier: `Exact · known pot/combos` + `Unavailable · unknown cards`
55	**Personal Strategy Construction vs. Baseline Mix** *Prereq:* #13, #15, #22	Authoring your own intended preflop and postflop gameplan (Personal Strategy) and comparing it descriptively (*inside mix, outside mix, close*) against Riverline's baseline without treating differences as errors (Brief §2.3 & §2.5).	Blindly copying a solver or heuristic chart you cannot execute postflop, or assuming any intentional exploitative simplification in your Personal Strategy is a "mistake."	In Riverline's Personal Strategy flow, declare your intended Cutoff RFI and vs-3-bet response for your 40 BB home game, then review descriptive comparison tags against the baseline.	**Judgement** Tier: `Estimated · descriptive comparison`

#	Concept & Prereqs	One-Line Definition (Original)	Typical Leak / Baseline Divergence	Quick Exercise Idea (Original)	Maths vs. Judgement & Riverline Tier
56	**Auditing Observed Play vs. Intended Strategy (Execution Drift)** *Prereq:* #48, #50, #55	Separating two distinct questions during post-session review: (1) Did I follow my own intended Personal Strategy? and (2) Should I update my Personal Strategy based on baseline/pool evidence? (Brief §2.5).	Silently rewriting your intended strategy after a hand ("I meant to call there") to justify an in-game tilt or boredom deviation.	Review 10 reconstructed hands from a session and tag each divergence as either *Execution Drift* (broke your own rule) or *Intentional Plan* (followed your rule, now evaluating the rule).	**Judgement** Tier: `Exact · plan match check` + `Estimated · baseline diff`
57	**Epistemic Confidence Tiers (Exact, Verified, Estimated, Unavailable)** *Prereq:* #4, #5, #11, #36	Distinguishing deterministic math (`Exact · scoped`), solver-benchmarked approximations (`Verified`), heuristic baselines (`Estimated`), and unmodeled live factors (`Unavailable`) during study (Brief §2.1).	Treating an `Estimated` multiway heuristic or a 1-combo solver frequency difference (52% vs. 48% bet) as absolute mathematical law while ignoring an `Exact` pot-odds price violation.	Inspect 6 numbers on a Riverline Review screen (Pot Odds %, Rule of 4 Equity %, 2-Bet Flop Heuristic Mix, Multiway 4-Way EV, Physical Tell Note) and assign the correct Confidence Tier to each.	**Hybrid** Tier: `Exact · epistemic taxonomy` (Brief §2.1)
58	**Off-Table Study Loops vs. Live Decision Boundaries** *Prereq:* #48, #50, #57	Using fast mental heuristics (pot odds benchmarks, SPR buckets, combo counts) internalized during off-table study while respecting strict ethical and legal bans on real-time assistance (Brief §2.6).	Attempting to look up charts or run calculators mid-hand during online play (violating site rules and Riverline's core policy) instead of logging the hand for post-session review.	Practice the 15-second "Between-Hands Tag": immediately after a live home-game hand ends, jot down only effective stack, position, board, and bet sizes to reconstruct after the session.	**Judgement** Tier: `Exact · policy boundary` (Brief §2.6)

5. Epistemic Confidence Architecture & Shared Domain Mapping

Principles 2.1 through 2.4 of `RIVERLINE_BRIEF.md` establish an industry-leading standard for analytical honesty: *"Every number says where it comes from… Poker maths live only in the shared domain; the UI formats, never computes."* To translate our 58-concept curriculum directly into engineering specifications for Milestone 1 (Trust Package) and Milestone 2 (Learn, Equity, Review, and Personal Strategy reworks), the table below defines how each category of concept maps to Riverline's four confidence tiers and to specific modules inside `shared/poker-domain/`.

Riverline Confidence Tier	Required Scope Badge Format in UI	Mapped Curriculum Concepts (#1–#58)	Canonical Owner in `shared/poker-domain/` & Verification Rule
1. Exact *Deterministic & Scoped*	`Exact · price only` `Exact · combinatorics` `Exact · given range` `Exact · stack geometry`	#3, #4, #5, #6 (exact outs), #10, #11, #16 (pot odds), #17, #18, #23, #24, #25, #27, #35, #36, #38, #49, #51 (SPR), #53 (compound prob)	Computed strictly in `shared/poker-domain/` (`potOdds.js`, `spr.js`, `combinatorics.js`, `equity.js`, `sizing.js`). Zero tolerance for approximation; unit-tested in Node's test runner.
2. Verified *Measured vs. Reference*	`Verified · vs exact (±1.8%)` `Verified · vs postflop-solver` `Verified · vs HRC Pro`	#6 (Rule of 4 & 2 shortcut), #37 (MDF vs. solver defense), #42, #45, and M3 solver-backed postflop/preflop spots (#13, #21, #22, #39, #40)	Heuristic or precomputed table stored in `shared/poker-domain/` and validated against exact combinatorial enumeration, b-inary/postflop-solver, or HRC Pro exports within a documented tolerance.
3. Estimated *Heuristic Baseline / Range Judgement*	`Estimated · heuristic baseline` `Estimated · villain range` `Estimated · realization R`	#1, #2 (EV), #7, #8, #9, #12, #13 (M2), #14, #15, #19, #20, #22 (M2), #26, #28, #29, #30, #31, #32, #33, #41, #43, #44, #47, #52, #55, #56	Generated by Riverline's heuristic baseline engine or user-supplied range assumptions. UI must use descriptive language (*inside mix, outside mix, close*) and never label deviation a "mistake."
4. Unavailable *Unmodeled or Missing Data*	`Unavailable · unknown cards` `Unavailable · multiway EV` `Unavailable · live tell / state`	#34 (4+ way exact GTO EV), #46 (physical tells), #48 (mental state), #54 (unrevealed hole cards or unrecorded bets in live reconstruction)	When inputs contain `Unknown` seats or unsolved multiway trees, `shared/poker-domain/` returns explicit `Unavailable` tokens (or exact min/max bounds) rather than inventing numbers or folding unknowns.

6. Suggested First 8 Lessons (Re-Engineered for Riverline's Dual Track & Workstation Loop)

With the context of `RIVERLINE_BRIEF.md`, the first 8 onboarding lessons in the Learn Hub are upgraded so that every lesson: (1) explicitly displays Riverline Confidence Tier badges on all numbers, (2) contrasts the owner's **30–60 BB rake-free home game** with **100 BB+ 9-max raked online play**, (3) links directly to a workstation loop (Hand Reconstruction, Equity, Review, or Personal Strategy), and (4) adheres to "calm surface, deep data" with one primary action per view.

Lesson 1: Seat Geometry & Effective Stack Ceilings (Home 40 BB vs. Online 100 BB+)

- **Mapped Concepts:** #2 (Table Positions), #3 (Effective Stack Size), #53 (9-Max vs. 6–8 Handed Geometry).
- **Epistemic Badges Shown:** `Exact · stack cap` (Effective Stack), `Exact · compound probability` (players left to act), `Estimated · positional EV`.
- **Core Lesson Summary (Original):** Synthesizing *Professional No-Limit Hold'em* and *The Grinder's Manual*, acting last postflop lets you price draws, bluff capped checks, and control pot size. Simultaneously, your maximum risk or reward in any pot is strictly capped by the smallest active stack (`min(Stack_Hero, Stack_Villain)`). In a 9-max online game, UTG must navigate 8 uncapped opponents, whereas in a 6–8 handed home game with mixed 30–60 BB stacks, identifying who covers whom before calling is paramount.

- **Workstation Drill ("calm surface, one primary action"):** A clean jade-accented table view displays an 8-handed home game (stacks: 32 BB, 48 BB, 55 BB, 110 BB) and a 9-max online table. The user clicks the single seat holding postflop relative position and types the exact effective stack in BB, powered by `shared/poker-domain/`.

Lesson 2: The Price of a Call (Expected Value & Scoped Pot Odds)

- **Mapped Concepts:** #4 (Expected Value), #5 (Pot Odds & Breakeven Required Equity), #57 (Epistemic Confidence Tiers).
- **Epistemic Badges Shown:** `Exact · price only` (Breakeven Equity %), `Exact · given equity` (Dollar/BB EV).
- **Core Lesson Summary (Original):** Grounded in David Sklansky's *The Theory of Poker* and Red Chip Poker *CORE 2.0*, every call is evaluated by comparing the price offered by the pot (`Call / (Pot + Bet + Call)`) against your chance of winning. Riverline introduces its core trust principle here: the price of a call is always labelled `Exact · price only` because it depends solely on visible chips in the pot, whereas your actual win probability on earlier streets often carries an `Estimated` or `Verified` badge.
- **Workstation Drill:** A hero answer card displays a single river bet (25%, 33%, 50%, 75%, or 100% pot). The user selects the exact breakeven equity percentage (16.7%, 20%, 25%, 30%, 33.3%) on a clean number tile bar; hovering reveals the underlying fraction formula.

Lesson 3: Counting Outs, the Rule of 4 & 2, and Discounting Redraws

- **Mapped Concepts:** #1 (Relative Hand Strength), #6 (Outs & Rule of 4 and 2), #7 (Dirty Outs & Redraw Discounting).
- **Epistemic Badges Shown:** `Exact · combinatorics` (True hypergeometric equity), `Verified · Rule 4/2 (±2.1%)`, `Estimated · dirty out weights`.
- **Core Lesson Summary (Original):** Drawing from *The Theory of Poker* and Jonathan Little's PokerCoaching.com Cash Game Masterclass, each clean out adds ~2.1% equity with one card to come and ~4% across two cards—provided the opponent does not bet the turn and charge a second price. This lesson explicitly demonstrates Riverline's `Verified` tier by showing side-by-side how the Rule of 4 & 2 tracks exact combinatorial math for 1–12 outs while diverging slightly on monster 15-out combo draws, and how paired or monotone boards require discounting dirty outs.
- **Workstation Drill:** Users inspect a flop hand with an open-ended straight draw plus overcard kicker, toggle dirty outs on a 52-card strip, and compare the `Verified` mental shortcut against `shared/poker-domain/`'s `Exact · combinatorics` output.

Lesson 4: Future Payoffs vs. Stack Depth Ceilings (Implied Odds at 40 BB vs. 100 BB)

- **Mapped Concepts:** #8 (Implied Odds & Reverse Implied Odds), #51 (30–60 BB Short/Mid-Stack Cash Geometry).
- **Epistemic Badges Shown:** `Exact · required future payoff`, `Estimated · villain payoff frequency`.
- **Core Lesson Summary (Original):** Adapted from *Professional No-Limit Hold'em* and Ed Miller's *The Course*, implied odds bridges the gap when direct pot odds fall short, but only if effective stacks are deep enough to supply the missing chips. In the owner's 40 BB home game, calling a 3.5 BB raise with `33` requires winning ~28 BB postflop just to break even on an 11.8% set draw—nearly the opponent's entire remaining 36.5 BB stack! Conversely, in a 100 BB+ 9-max game, the same call requires less than a third of the villain's stack.

- **Workstation Drill:** The user compares an identical pocket pair (`44`) facing a 3 BB open at 40 BB (home game) vs. 110 BB (online). `shared/poker-domain/` calculates the exact BB shortfall on the call, and the user judges whether the required postflop payoff is realistic given the effective stack ceiling.

Lesson 5: The 13×13 Range Grid, Combinatorics & Preserving Unknowns

- **Mapped Concepts:** #9 (Hand Ranges & 13×13 Matrix), #10 (Combinatorics & Card Removal), #54 (Reconstructing Incomplete Hands & Unknown Preservation).
- **Epistemic Badges Shown:** `Exact · combinatorics` (6/4/12 combo math), `Unavailable · unknown cards` (unrevealed seats).
- **Core Lesson Summary (Original):** Synthesizing Run It Once *From The Ground Up* and Michael Acevedo's *Modern Poker Theory*, offsuit hands carry 12 combinations, pocket pairs carry 6, and suited hands carry 4. Visible board and hole cards remove combinations deterministically (e.g., holding an Ace cuts villain's `AA` from 6 combos to 3 and `AK` from 16 to 12). Crucially, following Brief §2.2, when reconstructing a live home-game hand where a third player mucked unseen, Riverline keeps that seat explicitly tagged as `Unknown` rather than pretending they folded preflop.
- **Workstation Drill:** Given a flop of `As-Kh-7d` and hero's `Ah-Qd`, the user adjusts combo count tiles for `AA`, `KK`, `77`, and `AK`, witnessing how card removal shifts the value-to-bluff density in the Equity view.

Lesson 6: Building Your Personal Preflop Strategy (Home No-Rake vs. Online 9-Max)

- **Mapped Concepts:** #13 (RFI by Position), #14 (Isolating Limpers), #16 (BB Defense), #17 (Rake Adjustments), #52 (Rake-Free Home Shifts), #55 (Personal Strategy vs. Baseline Mix).
- **Epistemic Badges Shown:** `Exact · 0% vs 5% rake pot odds`, `Estimated · heuristic baseline` (with descriptive comparison: *inside mix* / *outside mix* / *close*).
- **Core Lesson Summary (Original):** Informed by the Upswing Poker Lab and Ed Miller's *The Course*, preflop strategy must adapt to table rules. In 9-max online cash with rake, early position opens are tight (~11% UTG) and non-blind cold-calling is restricted by the rake drop. In a 6–8 handed rake-free home game at 30–60 BB, zero rake makes Big Blind defense and in-position flatting with high-card hands (`KQs`, `QJs`, `AJo`) significantly more profitable, while limpers are isolated to play heads-up with initiative.
- **Workstation Drill:** Connects directly to Riverline's **Personal Strategy Question Flow (M2)**. The user defines their own intended opening and limper-isolation plan for both table profiles and sees descriptive feedback (*inside baseline mix* or *close*) with zero judgmental error scolding.

Lesson 7: Stack-to-Pot Ratio (SPR) & Multi-Street Commitment Architecture

- **Mapped Concepts:** #11 (Hand vs. Range Equity), #18 (SPR & Commitment Thresholds), #19 (Equity Realization), #38 (Geometric Bet Sizing).
- **Epistemic Badges Shown:** `Exact · SPR ratio`, `Exact · geometric sizing %`, `Estimated · commitment threshold`.
- **Core Lesson Summary (Original):** Grounded in *Professional No-Limit Hold'em* and Andrew Brokos's *Play Optimal Poker*, computing `SPR = Effective_Stack / Flop_Pot` the instant the flop falls dictates your entire multi-street plan. At SPR 1.5–4 (typical 3-bet pots at 100 BB or single-raised/3-bet pots in a 40 BB home game), top pair top kicker and overpairs are comfortable stack-off hands using geometric bet sizing. At SPR 12+ (single-raised 100 BB+ pots), one-pair hands require pot control on wet runouts.
- **Workstation Drill:** The user is dealt `Ah-Kd` on `Kc-8s-5d` across three SPR regimes (SPR 1.8 in a 40 BB 3-bet pot, SPR 5.0 in a 40 BB SRP, and SPR 14.5 in a 100 BB SRP) and selects the uniform geometric bet size computed by `shared/poker-domain/`.

Lesson 8: Reasons to Bet, C-Betting & Auditing Your Own Session

- **Mapped Concepts:** #12 (Reasons to Bet), #20 (Flop Textures), #22 (C-Betting IP vs. OOP), #23 (Bluff Breakeven Math), #56 (Auditing Observed Play vs. Intended Strategy).
- **Epistemic Badges Shown:** `Exact · bluff breakeven %` (`Bet / (Pot + Bet)`), `Estimated · heuristic baseline`, `Exact · Personal Strategy match`.
- **Core Lesson Summary (Original):** Synthesizing *The Grinder's Manual* and Matthew Janda's *Applications of No-Limit Hold'em*, every bet must have a clear primary purpose: getting called by worse hands (Value) or folding out better hands / denying equity (Bluff), calibrated against the exact bluff breakeven threshold (`Bet / (Pot + Bet)`). In post-session review, Riverline separates whether your bet matched your own intended **Personal Strategy** from how it compares to the heuristic baseline.
- **Workstation Drill:** The user replays 5 reconstructed hands from their own session (or a starter pack), declares their reason to bet (Value vs. Bluff + required fold %), and audits whether any divergence was *Execution Drift* or an *Intentional Plan*.

7. Technical Implementation, Trilingual Localization (EN/RU/HE) & Solver Licensing

7.1 Trilingual Terminology & Bidirectional Token Isolation (EN / RU / HE)

Section 1 of `RIVERLINE_BRIEF.md` mandates full support for **English (EN), Russian (RU), and Hebrew (HE)**, specifying *"Hebrew right-to-left with left-to-right poker notation."* In Chromium/Electron and mobile browsers, embedding ASCII poker notation (such as `AKo`, `J♠ T♠`, `3:1`, `SPR = 4.5`, or `+2.4 bb`) directly inside an RTL Hebrew sentence without bidirectional isolation causes the Unicode Bidirectional Algorithm (UBA) to invert card ranks, minus signs, and colons. All poker tokens rendered by the UI formatter must be wrapped in `<bdi dir="ltr">` or `<span dir="ltr" class="poker-token">`. The table below standardizes core curriculum terminology across all three languages:

Curriculum Concept / Term	English (EN)	Russian (RU — Кириллица)	Hebrew (HE — RTL + LTR Tokens)
Pot Odds & Required Equity	Pot Odds / Required Equity	Шансы банка / Требуемое эквити	סיכויי קופה (Pot Odds) / אקוויטי נדרש
Stack-to-Pot Ratio (SPR)	Stack-to-Pot Ratio (SPR)	Отношение стека к банку (SPR)	יחס ערימה לקופה (SPR)
Effective Stack Size	Effective Stack (`40 bb`)	Эффективный стек (`40 бб`)	ערימה אפקטיבית (40 bb)
Hand Range & Blockers	Hand Range / Blockers	Диапазон рук / Блокеры	טווח ידיים / חוסמים (Blockers)
Confidence Tiers (Brief §2.1)	Exact · Verified · Estimated · Unavailable	Точно · Проверено · Оценка · Недоступно	מדויק · מאומת · מוערך · לא זמין
Baseline Mix Comparison (§2.3)	Inside mix / Outside mix / Close	В миксе / Вне микса / Близко	בתוך התמהיל / מחוץ לתמהיל / קרוב

7.2 Solver Licensing Verification & M2 → M3 Precomputation Strategy

Section 5 of `RIVERLINE_BRIEF.md` notes current solver findings marked *"(to verify; not legal advice)"* and highlights a key gap: *"preflop and multiway for the owner's game types."* We verified the primary repositories and licensing agreements directly:

- **Postflop Heads-Up Verification (b-inary/postflop-solver):** Inspection of the official GitHub repository confirms that `b-inary/postflop-solver` (alongside `desktop-postflop` and `wasm-postflop`) is an open-source Rust postflop solver library released under the GNU Affero General Public License v3.0 (AGPL-3.0). Under copyright law and standard AGPL-3.0 interpretation, running an unmodified AGPL-3.0 binary offline on a local developer machine to generate static numerical strategy tables (JSON/binary equity and frequency lookups) does not subject the numerical output data itself to AGPL copyleft, provided no solver source code or WASM binary is linked or shipped inside the closed-source Riverline client. **[Judgement]** To maintain strict compliance with Brief Principle 2.7 ("record every outside component's licence before use"), Riverline should keep the offline solver generation scripts in a separate, isolated repository and ship only compact, precomputed numerical reference packs for the 58 curriculum drill spots.
- **Preflop 30–60 BB & 9-Max Verification (HoldemResources Calculator EULA):** Direct verification of the official HoldemResources Calculator (HRC) v4 End-User License Agreement (dated October 1, 2024) confirms the exact clause permitting commercial use of exported calculation results:

  > *"The Licensee may use the Product and its components only through the user interface accessible via the provided launcher. Sharing the license key or making the Product accessible to others is prohibited, especially through automated means... However, the Licensee is permitted to distribute, openly share, or use the results of authorized calculations for commercial purposes, provided these results were produced by the Licensee in compliance with this Agreement and applicable law."* (HoldemResources Calculator EULA)

  [Judgement] While Brief §2.7 notes that paid spending is currently deferred, HRC Pro directly solves the owner's exact preflop gap: it computes 6–8 handed 30–60 BB zero-rake preflop Nash approximations and 9-max 100 BB+ raked preflop ranges, and its EULA explicitly authorizes exporting and shipping those solved numerical charts in a commercial or free application. Until an HRC Pro subscription is authorized in M3, M2 preflop lessons should use Riverline's heuristic baseline clearly badged as `Estimated · heuristic baseline`.

8. Risks, Contradictions Flagged & Open Questions

Following Section 7 of `RIVERLINE_BRIEF.md` ("when something here seems wrong, say so instead of working around it... keep outputs structured: a short answer first, then the evidence table, then risks and open questions"), we highlight four architectural risks and open questions for the owner's decision:

Risk or Flagged Tension	Evidence & Technical Mechanism	Recommended Resolution / Question for Owner
1. Multiway Postflop Equilibrium Gap in Home Games	In 6–8 handed rake-free home games, 3-way to 5-way flops are the norm (Concept #34 & #52). However, multiway postflop Nash equilibrium is game-theoretically fragile (due to implicit collusion/coalition shifting) and unsupported by `b-inary/postflop-solver` (which is strictly heads-up).	**[Judgement]** Never badge multiway postflop strategy frequencies as `Verified`. Keep exact math (pot odds, compound probabilities, raw multiway equity on known ranges) badged `Exact`, and badge all multiway betting heuristics as `Estimated · multiway heuristic`.

Risk or Flagged Tension	Evidence & Technical Mechanism	Recommended Resolution / Question for Owner
2. Tension Between "No Mistake" Principle (§2.3) & Exact Math Drills	Brief §2.3 states: *"Heuristic disagreement is never a 'mistake'."* However, in pure math drills (e.g., Lesson 2: computing that a half-pot call requires `25.0%` equity), entering `33%` is a factual arithmetic error against an `Exact` domain calculation, not a heuristic disagreement.	**[Judgement]** Distinguish `Exact` math checks from `Estimated` strategy choices: for `Exact` math inputs, calmly show the exact formula difference without harsh buzzer sounds; for `Estimated` strategy lines, strictly use *inside mix / outside mix / close*. Confirm if owner agrees.
3. `b-inary/postflop-solver` Development Status	The official GitHub repository b-inary/postflop-solver is currently marked *"[Development suspended]"* by its author, though the Rust crate and CLI/GUI remain functional and stable for standard NLHE trees.	Since Riverline only needs offline batch solving to generate M3 verification tables, the suspended upstream status is low risk, but we should pin the exact commit hash and record its AGPL-3.0 license in Riverline's sourcing log.
4. Scope of First 8 Lessons in M2 Release	Building interactive widgets for all 58 concepts at once would delay Milestone 2. The First 8 Lessons cover 18 core concepts (#1–#14, #16–#20, #22–#23, #38, #51–#57) and connect all four M2 reworks (Personal Strategy, Reconstruction, Equity, Review).	**Open Question for Owner:** Should M2 Learn ship with the First 8 Lessons fully interactive (backed by `shared/poker-domain/`) and the remaining 40 concepts available as structured reference cards + sandbox links until M3?

9. Complete Bibliography & Verified Sources

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14. b-inary. *b-inary/postflop-solver: An efficient open-source postflop solver library written in Rust* (GitHub Repository, AGPL-3.0 License). Technical verification source for M3 offline postflop reference generation.
15. HoldemResources.net. *HoldemResources Calculator (HRC v4) End-User License Agreement* (October 1, 2024). Legal verification source confirming commercial distribution rights for exported preflop calculation results.